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Chromosome 20 RNA-seq Analysis: TDP-43 Knockout vs Wild-Type Rescue

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Abstract

We analyzed RNA-seq data from TDP-43 knockout (KO) and wild-type rescue (WT) HeLa cells, focusing on chromosome 20 genes. Using a standard bioinformatics pipeline (FastQC, fastp, Salmon, DESeq2), we identified **14 differentially expressed genes (DEGs)** on chr20 (adjusted $p < 0.05$, $|\log_2 FC| \geq 1$). Of these, 12 were upregulated and 2 were downregulated in KO cells. The upregulated genes included transcription factors (**FOXS1**, **MAFB**), signaling molecules (**SIRPA**, **PLCB1**), and ion channel **KCNG1**. The downregulated genes were **TSHZ2** (a transcriptional repressor) and **ENSG00000289720**. Gene Ontology enrichment analysis revealed significant terms related to interleukin-12 response, monocyte extravasation, and negative regulation of transcription. These results show that TDP-43 loss disrupts gene regulatory and immune-related networks even within a single chromosome.

Introduction

TDP-43 (encoded by *TARDBP*) is an RNA-binding protein involved in RNA splicing, stability, and transport. It normally resides in the nucleus, but in diseases like amyotrophic lateral sclerosis (ALS) and frontotemporal dementia (FTD), TDP-43 mislocalizes to the cytoplasm and forms toxic aggregates. Previous studies have shown that TDP-43 loss leads to widespread changes in gene expression and RNA splicing.

Most RNA-seq studies examine TDP-43 effects genome-wide. Here, we focused on chromosome 20 alone to test whether TDP-43's regulatory impact can be detected at a single-chromosome level. Chromosome 20 contains genes involved in transcription regulation, immune signaling, and other relevant processes, making it a good candidate for this focused analysis.

Our goals were to: (1) process RNA-seq data through a complete bioinformatics pipeline, (2) identify differentially expressed chr20 genes between KO and WT conditions, and (3) interpret the biological meaning of these changes.

Dataset

We used publicly available RNA-seq data from GEO (accession **GSE136366**). The experiment used HeLa cells with 3 TDP-43 KO replicates and 3 WT rescue replicates. Sequencing was

performed on an Illumina HiSeq 2500 (paired-end, 2×76 bp). Analysis was restricted to chromosome 20 using the GRCh38 reference genome (Ensembl release 115).

Methods

Quality Control and Trimming

Raw reads were checked with **FastQC** and summarized with **MultiQC**. All samples had consistent quality (~1.2–1.5 million read pairs, 76 bp reads, ~51% GC, 4–10% duplication by fastp). Adapter trimming and quality filtering were done with **fastp** (Q20 threshold, minimum 36 bp length).

Table 1. Raw Read Statistics

Sample ID	Condition	Pairs (M)	Read Len (bp)	%GC	%Dup	Map Rate
SRR10045016	KO	1.24	76	51	8.7	93.8%
SRR10045017	KO	1.27	76	51	8.5	92.2%
SRR10045018	KO	1.23	76	51	4.3	93.0%
SRR10045019	WT	1.38	76	51	9.5	93.6%
SRR10045020	WT	1.45	76	51	8.9	92.3%
SRR10045021	WT	1.29	76	51	5.2	92.3%

Quantification

Transcript abundances were quantified using **Salmon** (v1.5.2) against a chr20 transcriptome index. Mapping rates were consistently high (92–94%). Gene-level counts were obtained using **tximport**, producing a count matrix of 34,559 features across 6 samples.

Differential Expression Analysis

We used **DESeq2** (v1.38.0) to compare KO vs WT. DESeq2 normalizes for library size and applies the Wald test with Benjamini-Hochberg correction. We used lfcShrink for more reliable fold-change estimates. Genes were called significant at **adjusted p-value < 0.05** and **$|\log_2 FC| \geq 1$** (at least 2-fold change). Out of 365 genes that received adjusted p-values, 14 met these criteria.

Results

Quality Assessment

Library size comparison and count distribution boxplots confirmed consistent sequencing depth across all samples, with no major outliers.

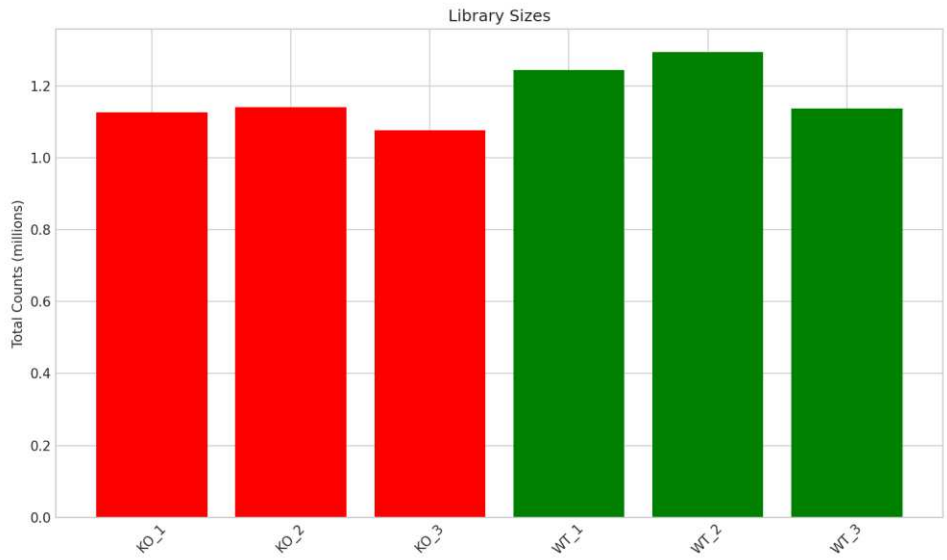

Figure 1. Library sizes. Bar plot showing total counts per sample. WT samples had slightly higher depth than KO, but this was accounted for by DESeq2 normalization.

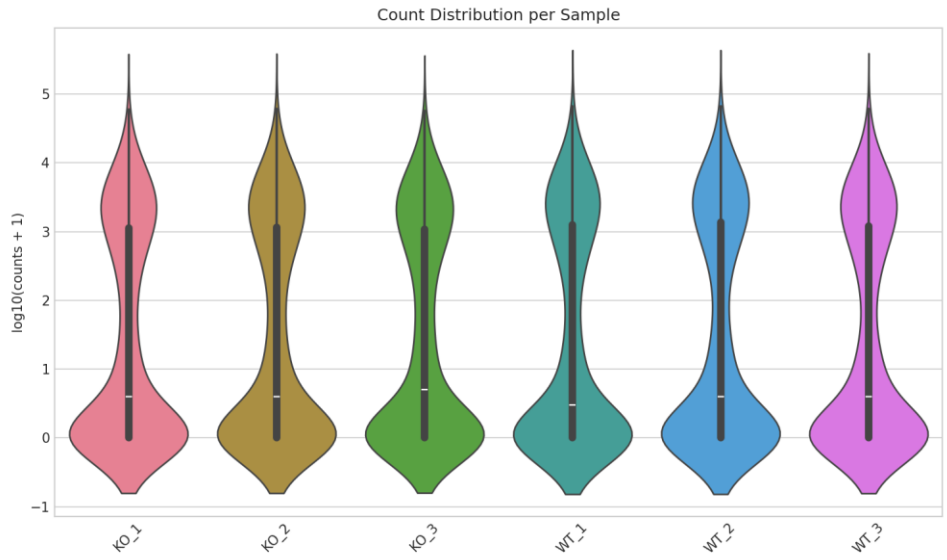

Figure 2. Count distribution boxplot. Log-transformed normalized counts show similar distributions across all six samples, confirming successful normalization.

Sample Clustering

PCA analysis showed clear separation between KO and WT samples along PC1 (50.4% of variance), confirming that the dominant source of variation is the experimental condition rather than technical noise. The sample correlation heatmap further supports this grouping.

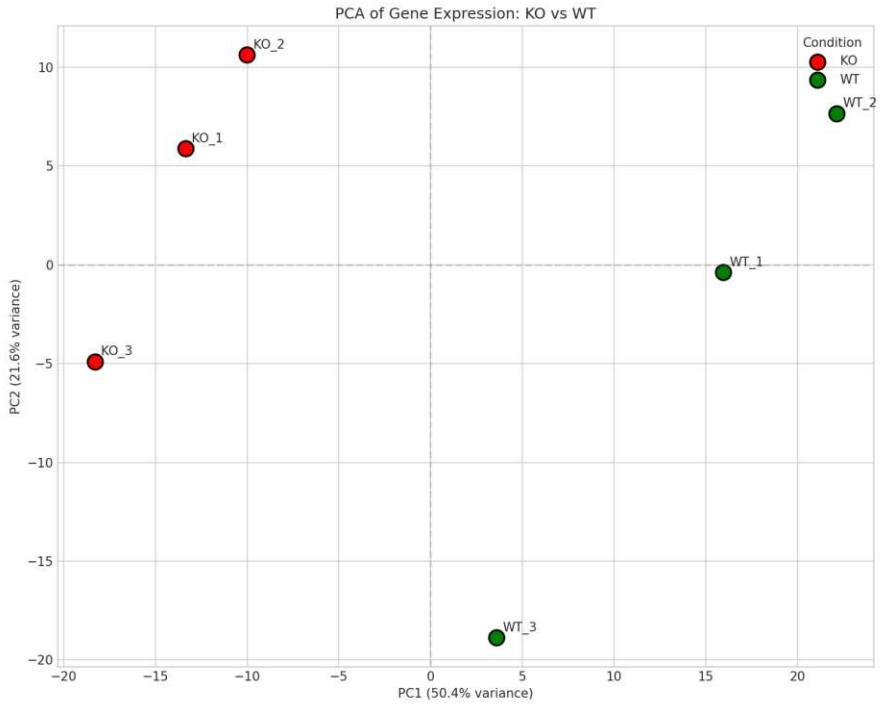

Figure 3. PCA plot. KO samples (red) cluster on the left and WT samples (green) on the right along PC1, which explains 50.4% of the total variance. PC2 (21.6%) captures within-group variability.

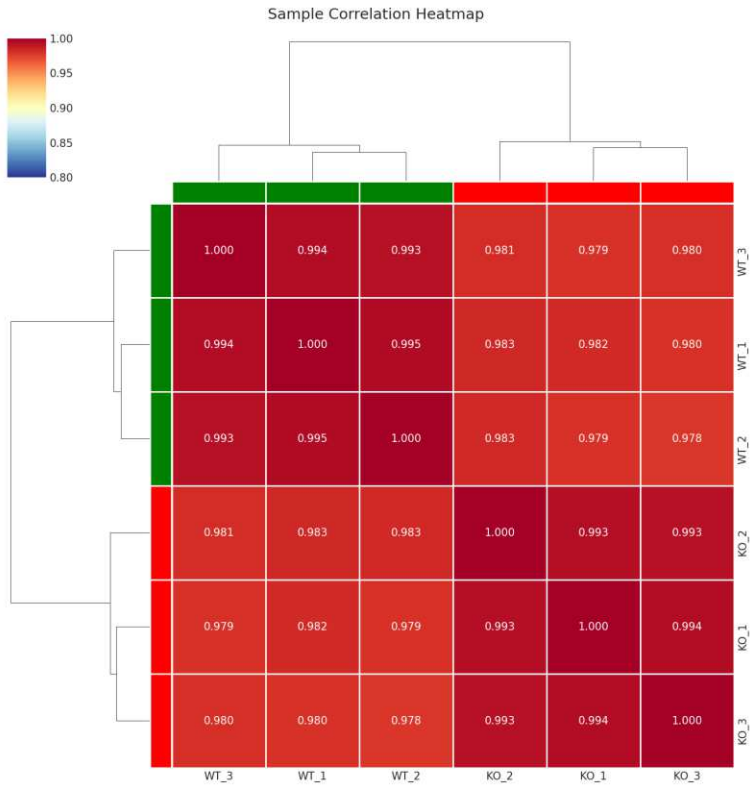

Figure 4. Sample correlation heatmap. Pairwise Pearson correlations of normalized gene expression. Samples cluster by condition (KO vs WT).

Differentially Expressed Genes

Out of 365 chr20 genes tested, **14 genes** were significantly differentially expressed (~3.8%). Of these, **12 were upregulated** and **2 were downregulated** in KO cells.

Table 2. Top Upregulated Genes in TDP-43 KO (Chr20)

Gene	Description	$\log_2$FC	Adj. p-value
FOXS1	Forkhead Box S1 (transcription factor)	+4.591	6.38×10^{-8}
PKIG	Protein kinase inhibitor gamma	+3.464	1.44×10^{-22}
MAFB	MAF transcription factor B	+2.411	5.53×10^{-6}
SIRPA	Signal-regulatory protein alpha	+2.315	7.02×10^{-25}
KCNG1	Potassium channel subfamily G member 1	+2.102	1.21×10^{-9}

Table 3. Downregulated Genes in TDP-43 KO (Chr20)

Gene	Description	$\log_2$FC	Adj. p-value
TSHZ2	Teashirt zinc finger homeobox 2	–1.944	4.00×10^{-28}
ENSG00000289720	Uncharacterized lncRNA	–1.181	8.08×10^{-4}

MA Plot

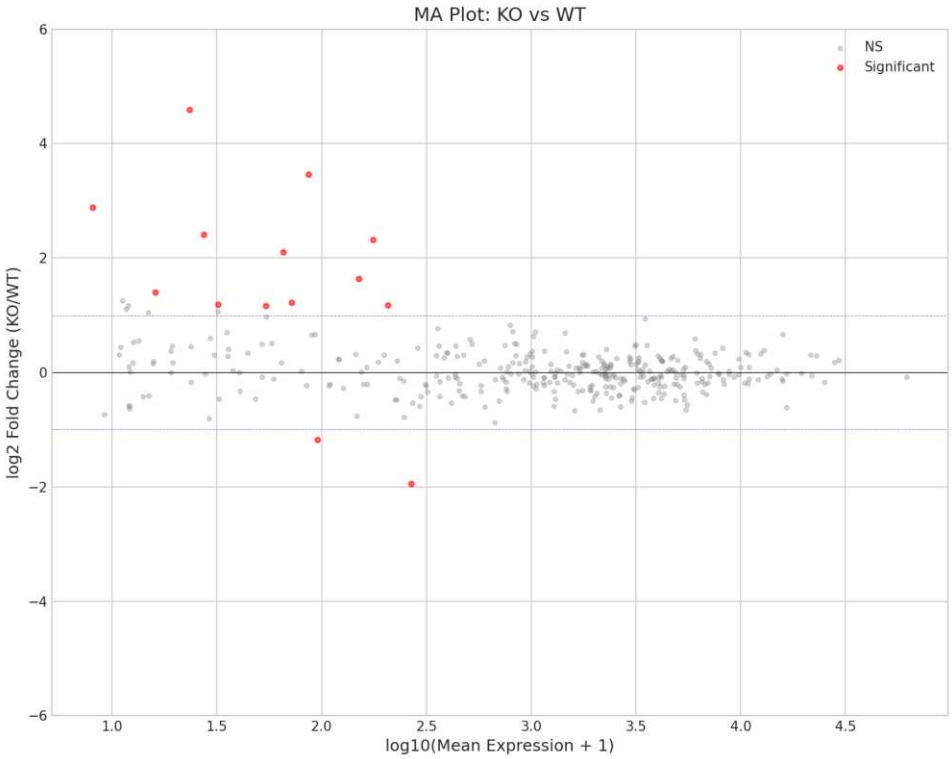

Figure 5. MA plot. Each dot is a chr20 gene. X-axis: mean expression (log10), Y-axis: $\log_2$ fold change. Red dots are significant DEGs. Most genes cluster around zero, while 14 genes show significant changes. The dashed lines mark the ±1 $\log_2$FC cutoff.

Volcano Plot

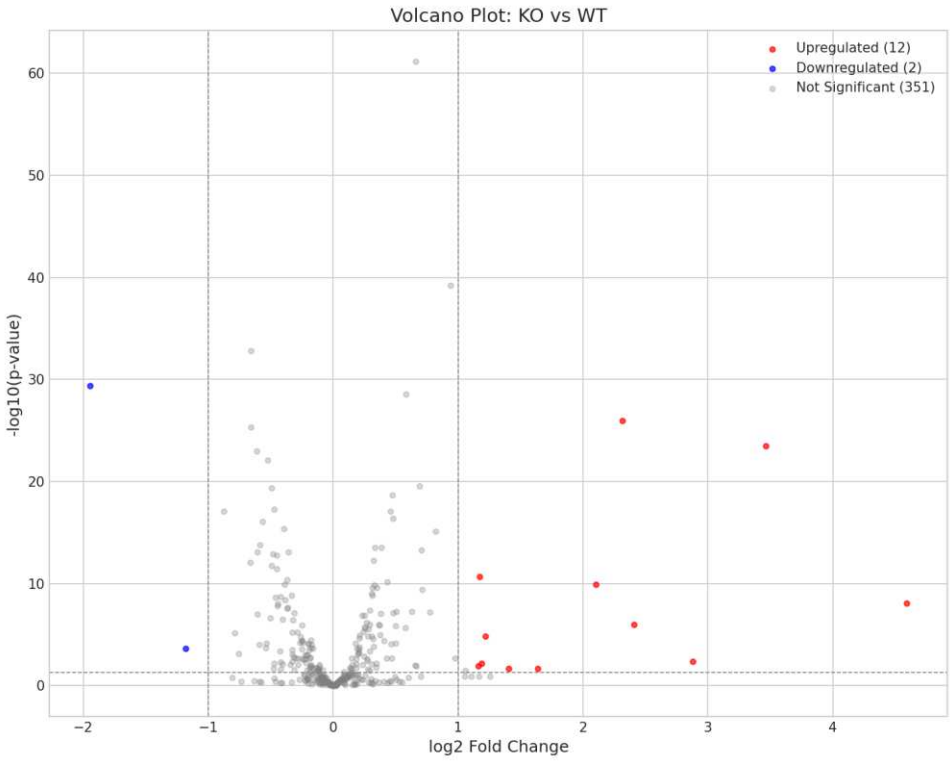

Figure 6. Volcano plot. Red dots: 12 upregulated genes; blue dots: 2 downregulated genes; gray: not significant (351 genes). TSHZ2 is the most significant downregulated gene (padj ~10^{-28}). SIRPA and PKIG are among the most significant upregulated genes.

Top DEGs Heatmap

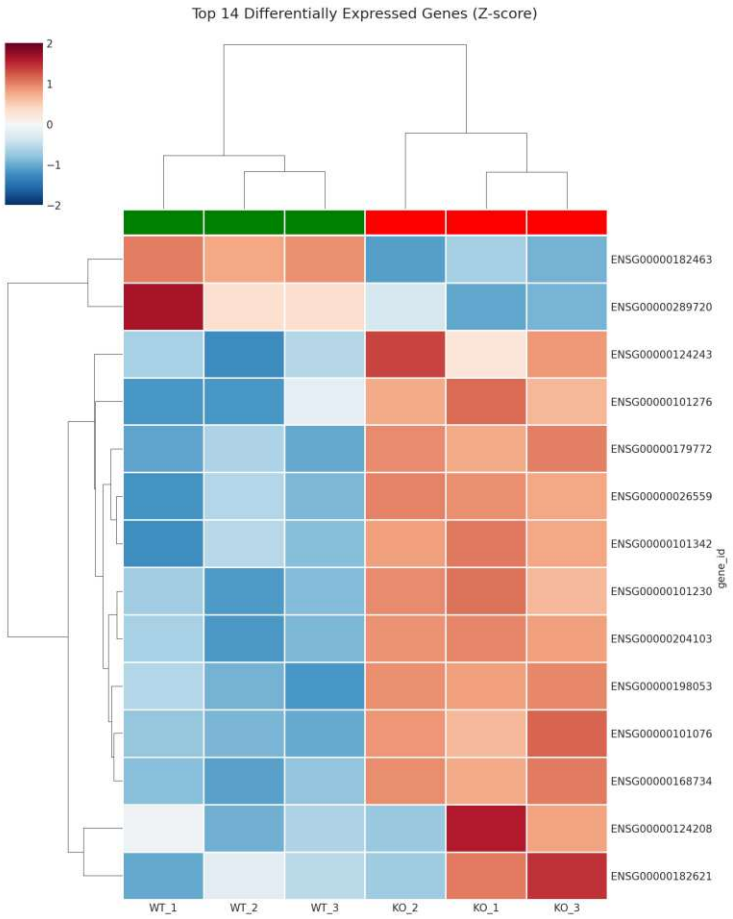

Figure 7. Heatmap of top 14 DEGs. Z-score normalized expression across all samples. The hierarchical clustering clearly separates WT (left, green) from KO (right, red) samples. The two downregulated genes (TSHZ2, ENSG00000289720) show opposite patterns to the 12 upregulated genes.

Gene Ontology Enrichment

Unlike the previous analysis iteration, this updated analysis found **significant GO enrichment results**. Using g:Profiler, we found 20 significant GO Biological Process terms for all DEGs, and 68 terms for the upregulated genes alone. No significant KEGG pathways were found. The top enriched terms fell into three main categories:

Immune/inflammatory response: The most significant terms were cellular response to interleukin-12 ($p = 0.006$), monocyte extravasation ($p = 0.004$), and cellular extravasation ($p = 0.048$). These involved SIRPA and PLCB1, both of which participate in immune cell signaling.

Transcriptional regulation: Several terms related to negative regulation of DNA-templated transcription (p = 0.027) and negative regulation of RNA metabolic process (p = 0.027) were enriched, involving 5 of our DEGs (PKIG, FOXS1, MAFB, HNF4A, PLCB1).

NF-κB signaling: Negative regulation of I-kappaB phosphorylation (p = 0.027) was enriched, driven by SIRPA. This is relevant because TDP-43 pathology is known to activate NF-κB-related stress pathways.

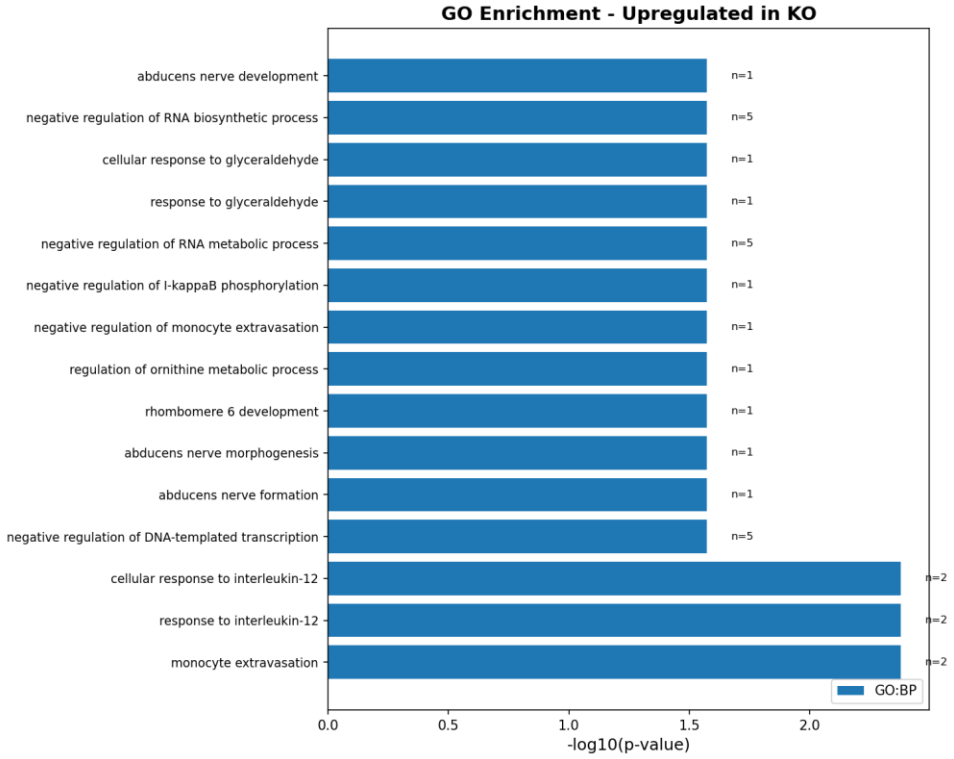

Figure 8. GO enrichment bar plot (upregulated genes). Top enriched Biological Process terms for the 12 upregulated DEGs. The most significant terms relate to interleukin-12 response and monocyte extravasation. The number of DEGs in each term is shown (n=).

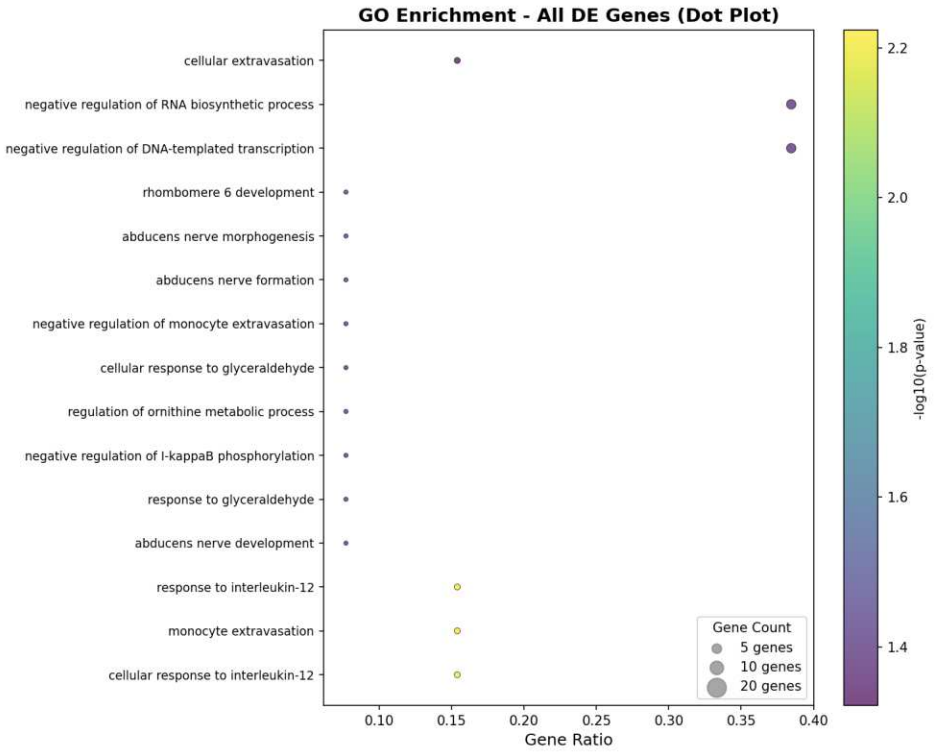

Figure 9. GO enrichment dot plot (all DEGs). Dot size represents the number of genes in each term, and color indicates statistical significance.

Discussion

Our analysis identified 14 significantly differentially expressed genes on chromosome 20 following TDP-43 knockout. The main finding is that TDP-43 loss predominantly causes upregulation (12 out of 14 DEGs), suggesting that TDP-43 normally helps keep certain genes in check.

Several upregulated genes are transcription factors (FOXS1, MAFB, HNF4A), which could mean cells are activating backup regulatory programs to compensate for TDP-43 loss. The strong upregulation of SIRPA and PLCB1 — both involved in immune signaling — along with the enrichment of interleukin-12 and monocyte extravasation GO terms, suggests that TDP-43 loss triggers an immune-like stress response even in non-immune cells like HeLa. KCNG1, an ion channel subunit, was also notably upregulated.

TSHZ2 and ENSG00000289720 were the two downregulated genes. TSHZ2 is a transcriptional repressor, so its decrease could lead to de-repression of its target genes. The second gene

(ENSG00000289720) appears to be an uncharacterized lncRNA whose function is not yet known.

Compared to the initial analysis, this updated run identified one additional DEG and, importantly, produced significant GO enrichment results. The enrichment of transcription-related and immune-response terms aligns well with TDP-43's known roles in RNA metabolism and the cellular stress responses observed in TDP-43 proteinopathies.

Limitations

We only analyzed chromosome 20, missing genome-wide effects. The HeLa cell line may not fully represent neuronal TDP-43 biology relevant to ALS/FTD. With 3 replicates per condition and moderate sequencing depth, subtle changes may have been missed. We focused on gene-level expression and did not examine alternative splicing, which is one of TDP-43's major regulatory functions.

Conclusion

TDP-43 knockout causes detectable gene expression changes on chromosome 20, with 14 DEGs (mostly upregulated transcription factors and immune-related genes). The significant GO enrichment results point to disrupted transcriptional regulation and inflammatory signaling. These findings support TDP-43's role as a key regulator of gene expression and provide a chromosome-level view of its impact.

Tools and Reproducibility

Software: FastQC (v0.11.9), fastp (v0.23.2), Salmon (v1.5.2), DESeq2 (v1.38.0), tximport, g:Profiler/clusterProfiler (v4.2.2), MultiQC (v1.11), R (v4.2).

GitHub: https://github.com/ProgAm1/TDP43_chr20_RNAseq

Data source: GEO accession GSE136366

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Declarations

Data Availability

The RNA-seq datasets analyzed in this study are publicly available from the NCBI Gene Expression Omnibus (GEO) under accession number GSE136366 (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE136366). The GRCh38 reference genome and Ensembl release 115 annotations used for read mapping are available from the Ensembl genome browser (https://www.ensembl.org).

Code Availability

All analysis scripts and code used in this study are publicly available on GitHub at https://github.com/ProgAm1/TDP43_chr20_RNAseq.

Competing Interests

The authors declare no competing interests.

Funding

No specific funding was received for this work.

Ethics Statement

This study used only publicly available, de-identified RNA-seq data from the NCBI Gene Expression Omnibus (GEO accession GSE136366). No new human or animal experiments were conducted. Ethical approval was therefore not required.

AI Disclosure

An AI language model (Claude, Anthropic) was used to assist with manuscript drafting and editing. The AI was not listed as an author. All scientific content, analysis, interpretation, and final decisions regarding the manuscript were made by the human authors, who take full responsibility for the work.